Validation Analysis of a Real-Data Explainable AI Framework for Intrusion Detection Systems

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1 Context

Explainable Artificial Intelligence (XAI) has become a critical requirement for deploying machine learning models in critical domains such as cybersecurity. In Intrusion Detection Systems (IDS), explanations are not merely auxiliary tools for post-hoc interpretability; they play a central role in operational decision making, forensic analysis, model auditing, and in establishing trust between automated detection systems and human experts [2].
Despite their widespread adoption, most state-of-the-art local post-hoc XAI methods, such as LIME [6], LEMNA [8], SHAP [7], and Integrated Gradients [9], rely on synthetic perturbations of input features to approximate local model behavior. When applied to network traffic data, these perturbation strategies often generate unrealistic or infeasible samples that violate protocol specifications, feature dependencies, or physical and statistical constraints inherent to real network communications. As a result, explanations may be derived from regions of the feature space that do not correspond to any realizable network behavior, undermining their reliability and limiting their operational relevance in cybersecurity settings.
To overcome these limitations, we propose a synthetic-free, real data–driven local explanation algorithm specifically designed for tabular cybersecurity data. The method constructs feature-specific local neighborhoods exclusively from real instances observed in the dataset, ensuring that all explanatory samples inherently respect domain constraints, and introduces stability-based metrics to quantify feature importance by measuring the consistency of model predictions within these neighborhoods.
Beyond interpretability, XAI methods can also be exploited to guide the generation of adversarial examples [1] [3][4][5], which are crafted inputs to exploit an IDS decision boundary in order to evade detection or trigger false alarms. Paradoxically, the effectiveness of such adversarial examples can itself serve as an indicator of the faithfulness of an explanation method to the underlying model behavior[15]. However, most existing adversarial techniques operate purely in feature space and frequently produce samples that are invalid in problem space, corresponding to non-realizable network connections[16].
The objective of this internship is therefore to rigorously validate and extend the proposed XAI framework, with an emphasis on explanation robustness and adversarial analysis under realistic constraints, with the objective to exploit these results to develop a visual analytics framework that supports human experts.

2 Overview of the work

The main tasks of the intern are:

- State-of-the-art analysis of XAI methods and their applications to cybersecurity:
  - Local and global XAI methods (e.g., LIME, SHAP, IG, counterfactual explanations.).
  - Applications of XAI in IDS
  - Existing robustness, faithfulness, and stability metrics for explanations.
- Robustness evaluation of the proposed explanation method across multiple IDS datasets:
  - Testing explanation stability across attack types
  - Comparing explanation behavior across correct and incorrect predictions (TP, TN, FP, FN).
  - Investigate metrics to assess neighborhood quality.
- Development of explanation driven adversarial examples:

- Systematic study of classical adversarial generation methods (FGSM [11], BIM [12], DeepFool [14], C&W [13]) to explainability-driven approaches.
- Implementing adversarial examples that are consistent with network traffic semantics.

3 Expected outcomes

At the end of the internship, we expect to provide:

- A validated and stress-tested XAI framework for IDS.
- A standardized evaluation framework for assessing and comparing XAI methods.
- An approach to generate adversarial examples that are consistent in problem space.
- Metrics to evaluate the quality of local neighborhoods used in XAI.

4 Desired profile of the intern

- Masters in cybersecurity, data science or mathematics
- Strong background in statistics and machine learning
- Strong background in python
- Familiarity with cybersecurity concepts and IDS datasets
- Interest in Explainable AI

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